

Integral 2010, Set 1

1. Give a necessary and sufficient condition for the existence of a maximum matching in a graph (+proof). **(13 p)**
2. Apply Dijkstra's algorithm to the graph in Figure 1. Explain the choice at each step. Which problem solves this algorithm? **(12 p)**
3. Explain what is a matching, a maximum matching and a complete matching in a graph. **(6 p)**
4. Prove that, when a BFS algorithm is applied to a connected graph, the end-points of each non-tree edge are either at the same level, or at consecutive levels. **(7 p)**
5. Find a binary code of minimum weighted length for encoding the following text *matematica e numai pentru matematicieni*. **(10 p)**
6. We have two urns. The first contains 6 white balls and 7 black balls and the second contains 10 white balls and 5 black balls. We extract a ball from the first urn and introduce it into the second one, then we extract from the second urn 5 balls, without reintroducing them back after each extraction. Let X denote the number of white balls extracted from the second urn. Calculate $E(X)$. **(8p)**
7. Which is the number of ordered selections without repetition of n objects from an m -set? (+argumentation). **(5p)**
8. The weak law of large numbers (definition). **(5p)**
9. The telephone lines serving an airline reservation office are all busy about 65% of the times. Let X be the random variable representing the number of calls necessary for connection. a) Which are the possible values of X ? Write the distribution of the random variable X . Which distribution has this r.v.?; b) Which is the expected number of tries for succeeding a reservation? c) Write the probability to perform at most 3 attempts; d) Write the probability to perform at least 2 attempts. **(8 p)**
10. How many positive solutions has the equation $x_1 + x_2 + x_3 + x_4 + x_5 = 16$? The same question, if in addition we require that $x_1 \geq 3$ and $x_4 \geq 5$. **(6p)**

Total: